Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/23/2018
PI Name:	Ajaev, Vladimir Ph.D.		
Project Title:	Modeling Phase Change and Thermocapillary Flow in the Constrained Vapor Bubble Experiment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75275-0156	**Congressional District:**	32
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015-16 Physical Sciences NNH15ZTT001N-15PSI-C: Use of the NASA Physical Sciences Informatics System – Appendix C
Start Date:	12/22/2017	**End Date:**	12/21/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Plawsky, Joel Sc.D. (Rensselaer Polytechnic Institute)		
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Performance Goal Text:			
	We propose theoretical and numerical studies of constrained vapor-liquid interfaces in the presence of phase change, i.e., evaporation and/or condensation. These studies will result in better understanding of heat and mass transfer in two-phase systems in which the effects of capillary forces and thermocapillarity are important. The main focus of the work is on a specific geometric configuration used in a recent space experiment, referred to as a Constrained Vapor Bubble (CVB). We propose to use asymptotic methods to simplify the system of governing equations for fluid flow and heat transfer for this configuration, followed by numerical solutions of the resulting equations using modern computational techniques. The proposed work has a wide range of ground and space applications including two-phase cooling systems for traditional electronic devices such as laptop computers as well as novel thermal management systems for emerging applications such as proton exchange membrane fuel cells and photovoltaic cells for solar energy		

Task Description:	harvesting. The small length scales in many of these systems imply that interfacial phenomena are very important while the effects of gravity are secondary. Some aspects of the proposed work are also relevant for modeling of physical processes such as self-assembly of colloidal particles from evaporating droplets, condensation of droplets in novel devices for energy conversion, and water desalination. The liquid transport mechanisms by capillary forces are also important to study for space applications, such as advanced cooling systems for spacecraft, since conventional methods for liquid transport can fail or become impractical in such applications. For example, any system that relies on the force of gravity for functioning is likely to fail in the microgravity environment. Transporting liquids by external pressure gradients, while not impossible in space, typically requires mechanical pumps, often making such systems impractical because of the payload weight restrictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed work has a wide range of ground and space applications including two-phase cooling systems for traditional electronic devices such as laptop computers as well as novel thermal management systems for emerging applications such as proton exchange membrane fuel cells and photovoltaic cells for solar energy harvesting.
Task Progress:	The following has been accomplished during the first year of the project: 1. The graduate student at Southern Methodist University (SMU) started working on the project in February, 2018. He has reviewed the literature on the topic of modeling of moving contact lines in the presence of evaporation, as well as closely related literature on CVB (constrained vapor bubble) experimental set-up and on heat pipes. He then went on to formulate a model for evaporation near contact lines in partially wetting systems in the configuration of a sessile axisymmetric droplet, motivated by observations of evaporating and condensing droplets in the CVB cuvette. The novel aspect of the formulation is the model of partial wetting not previously described in the literature on droplets in pure vapor environment. 2. The mathematical model was implemented numerically using a modification of several in-house codes together with writing several new subroutines as required by the new component of the formulation. The code was validated and used to observe the behavior of contact lines in the presence of evaporation. Local evaporation rates were found to decrease with the increase of the contact angle. The effect of Marangoni stresses has then been added to the model and simulations of the accordingly modified model are currently ongoing. 3. Motivated by the experimental observations of liquid transport in the corners of the CVB cuvette in microgravity, the graduate student started working on wedge flow models. The previously developed model of capillary-driven flow in a wedge was modified to include two new components: the pressure and temperature variation in the longitudinal direction. The numerical implementation of the model is currently underway. 4. During the course of this year we have also been vigorously working on reanalyzing the data from both CVB-2 and CVB-1 flight experiments and investigating how we can enhance our fringe analysis technique to extract even more information from the data collected. We are now trying to expand models that we developed for meniscus evaporation to partially wetting fluids and fluid mixtures. We have also begun developing 3-D, Phase-Field models of the CVB in preparation for trying to simulate the Marangoni fingers we observed in the system and the central drop.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Abstracts for Journals and Proceedings	Barrett J, Ajaev VS. "The effect of electrostatic forces on the evaporation of liquid droplets." 71st Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Atlanta, Georgia, November 18-20, 2018. Bulletin of the American Physical Society. 2018;Abstract: M13.00004. http://meetings.aps.org/Meeting/DFD18/Session/M13.4 ; accessed 10/30/18. , Oct-2018
Abstracts for Journals and Proceedings	Plawsky JL. "Micro-jet Formation at the Confluence of Marangoni and Capillary Flows." ASME 2018 16th International Conference on Nanochannels, Microchannels and Minichannels (ICNMM2018), Dubrovnik, Croatia, June 10-13, 2018. ASME 2018 16th International Conference on Nanochannels, Microchannels and Minichannels (ICNMM2018), Dubrovnik, Croatia, June 10-13, 2018. , Jun-2018
Abstracts for Journals and Proceedings	Ajaev VS, Gatapova EY. "Contact line motion on heated surfaces with spatially non-uniform properties." 71st Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Atlanta, Georgia, November 18-20, 2018. Bulletin of the American Physical Society. 2018;A10.00008. http://meetings.aps.org/Meeting/DFD18/Session/A10.8 ; accessed 10/30/18. , Oct-2018